COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212/Automating Server MAnagement	**SECTION:**	CPE31S2
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ACTIVITY NO. 11.1 **HOA 11.1: Containerization**

1. PROCEDURE OUTPUT

1. Create a new repository for this activity.

2. Install Docker and enable the docker socket.

3. Add to Docker group to your current user.

4. Create a Dockerfile to install web and DB server.

5. Install and build the Dockerfile using Ansible.

6. Add, commit and push it to your repository.

Step 1: Create a new repository for this activity

The document makes a folder for the activity inside the repo. You do the same.

1.1 Update your repo from GitHub so the push goes through later.

```
wdjavillonar@workstation:~$ cd ~/CPE212_Javillonar && git pull origin main
remote: Enumerating objects: 227, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 227 (delta 2), reused 1 (delta 1), pack-reused 216 (from 1)
Receiving objects: 100% (227/227), 20.44 MiB | 4.41 MiB/s, done.
Resolving deltas: 100% (81/81), completed with 1 local object.
From https://github.com/MadhelKathlynAnnJavillonar/CPE212_Javillonar
 * branch            main       -> FETCH_HEAD
   c5b8a86..f9b7863  main       -> origin/main
Updating c5b8a86..f9b7863
Fast-forward
 .../Javillonar_1.1 Hands-on Activity - Creating Virtual Machines  (1).pdf       | Bin 0 -> 2918711 bytes
 HOA1.1_Creating_Virtual_Machines/README.md                                      |   1 +
 HOA10.1_Log_Monitoring/README.md                                                |   1 +
 HOA123                                                                          |   1 +
 .../HOA_Familiarization for the new normal Lab_Javillonar.pdf                   | Bin 0 -> 3158675 bytes
 HOA1_Familiarization_New_Normal/README.md                                       |   1 +
 ...r_2.1 Hands-on Activity - SSH Key-Based Authentication and GIT Setup (1).pdf | Bin 0 -> 1046148 bytes
 HOA2.1_SSH_Key_Auth_and_Git_Setup/README.md                                     |   1 +
 ...lonar_3.1 Hands-on Activity - Install SSH server on CentOS or RHEL 8 (1).pdf | Bin 0 -> 1484156 bytes
 HOA3.1_Install_SSH_Server_CentOS_RHEL8/README.md                                |   1 +
 HOA4.1_Ansible_Basics/4.1 HOA_ Ansible Basics - Javillonar (1).pdf              | Bin 0 -> 1897572 bytes
 HOA4.1_Ansible_Basics/README.md                                                 |   1 +
 HOA4.1_Ansible_Basics/install_apache.yml                                        |  17 +++++++
 .../HOA-5.1-Implement Ansible roles in playbooks.pdf                            | Bin 0 -> 3667536 bytes
 HOA5.1_Ansible_Roles_in_Playbooks/README.md                                     |   1 +
 HOA5.1_Ansible_Roles_in_Playbooks/install_apache.yml                            |  13 +++++
 HOA5.1_Ansible_Roles_in_Playbooks/inventory                                     |   4 ++
```

1.2 Create the activity folder and go inside.

```
wdjavillonar@workstation:~/CPE212_Javillonar$ mkdir -p HOA11.1_Docker_Web_DB && cd HOA11.1_Docker_Web_DB
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$
```

1.3 Check the repo folder and your location.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ ls ~ && pwd
ansible-lab         CPE232_Javillonar  Documents  inventory  Pictures  snap       Videos
CPE212_Javillonar   Desktop            Downloads  Music      Public    Templates
/home/wdjavillonar/CPE212_Javillonar/HOA11.1_Docker_Web_DB
```

Step 2: Install Docker and enable the docker socke

The page you gave says to set up Docker's package repository first, so that is done before the install.

2.1 Install the tools needed to download from Docker's website.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo apt update && sudo apt install -y ca-certificates cu
rl
[sudo] password for wdjavillonar:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
144 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20260601~24.04.1).
ca-certificates set to manually installed.
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libcurl3t64-gnutls libcurl4t64
The following NEW packages will be installed:
  curl
The following packages will be upgraded:
  libcurl3t64-gnutls libcurl4t64
2 upgraded, 1 newly installed, 0 to remove and 142 not upgraded.
Need to get 906 kB of archives.
After this operation, 536 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64 amd64 8.5.0-2ubuntu10.15 [343 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.15 [227 kB]
37% [2 curl 5,388 B/227 kB 2%]
```

2.2 Create the folder where Ubuntu keeps repository keys.
2.3 Download Docker's official key. Ubuntu uses it to confirm that the packages come from Docker.
2.4 Make the key readable.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo install -m 0755 -d /etc/apt/keyrings
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/
gpg -o /etc/apt/keyrings/docker.asc
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo chmod a+r /etc/apt/keyrings/docker.asc
```

2.5 Add Docker's repository. It detects your Ubuntu version by itself.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/e
tc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION
_CODENAME}") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$
```

2.6 Refresh the package list so Ubuntu sees the new repository.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo apt update
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:2 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [69.0 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 117 kB in 2s (67.4 kB/s)
Reading package lists... 5%
```

2.7 Install Docker. This is the same install command shown in the document.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo apt install docker-ce docker-ce-cli containerd.io do
cker-buildx-plugin docker-compose-plugin -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  docker-ce-rootless-extras pigz
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin pigz
0 upgraded, 7 newly installed, 0 to remove and 142 not upgraded.
Need to get 102 MB of archives.
After this operation, 388 MB of additional disk space will be used.
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 2.4.1-2~ubuntu.24.04~noble [23.3 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
17% [1 containerd.io 18.5 MB/23.3 MB 79%]
```

2.8 Enable the Docker service and socket so they start now and on every boot.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo systemctl enable --now docker.service docker.socket
containerd.service
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
```

The socket is the channel the docker command uses to talk to the Docker engine.

2.9 Check that Docker is running.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-10-12 20:03:21 PST; 5min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 15514 (dockerd)
      Tasks: 9
     Memory: 40.0M (peak: 40.2M)
        CPU: 912ms
     CGroup: /system.slice/docker.service
             └─15514 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 12 20:03:20 workstation dockerd[15514]: time="2026-10-12T20:03:20.689307513+08:00" level=info msg="Restoring con… start."
Oct 12 20:03:20 workstation dockerd[15514]: time="2026-10-12T20:03:20.723091348+08:00" level=info msg="Deleting nfta…tatus 1"
Oct 12 20:03:20 workstation dockerd[15514]: time="2026-10-12T20:03:20.734617420+08:00" level=info msg="Deleting nfta…tatus 1"
Oct 12 20:03:21 workstation dockerd[15514]: time="2026-10-12T20:03:21.431114717+08:00" level=info msg="Loading conta…: done."
Oct 12 20:03:21 workstation dockerd[15514]: time="2026-10-12T20:03:21.455920676+08:00" level=info msg="Docker daemon…n=29.9.0
Oct 12 20:03:21 workstation dockerd[15514]: time="2026-10-12T20:03:21.456115442+08:00" level=info msg="Initializing buildkit"
Oct 12 20:03:21 workstation dockerd[15514]: time="2026-10-12T20:03:21.632219784+08:00" level=info msg="Completed bui…ization"
Oct 12 20:03:21 workstation dockerd[15514]: time="2026-10-12T20:03:21.635496023+08:00" level=info msg="Daemon has co…ization"
Oct 12 20:03:21 workstation dockerd[15514]: time="2026-10-12T20:03:21.635535604+08:00" level=info msg="API listen on…er.sock"
Oct 12 20:03:21 workstation systemd[1]: Started docker.service - Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
```

2.10 Test the installation, as in the document.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5e23090353324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

“Hello from Docker!” means the installation works.

Step 3: Add the Docker group to your current user

So Right now Docker only works with sudo and step lets the user run it directly, which the Ansible playbook needs later.

3.1 Create the group if it doesn’t exist yet.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo groupadd -f docker
```

3.2 Add user to the group.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo usermod -aG docker wdjavillonar
```

-a means add, so your other groups stay.
-G docker is the group you are joining.

3.3 Apply the change to your current terminal.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ newgrp docker
```

A group change normally applies only after logging in again. newgrp opens a new shell in the same terminal with the group already active. Keep using this terminal for the rest of the activityand If you close it, run newgrp docker again.

3.4 Test Docker without sudo.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

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For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

It should print “Hello from Docker!” with no password prompt. It won’t show the “Pulling from library” lines because the image was downloaded in Step 2.10.

Step 4: Create a Dockerfile to install the web and DB server

4.1 Make sure its in the activity folder.
cd ~/CPE212_Javillonar/HOA11.1_Docker_Web_DB

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$
```

4.2 Create the Dockerfile.
What each line does:
FROM ubuntu:22.04 starts from a clean Ubuntu image.
ENV DEBIAN_FRONTEND=noninteractive stops install questions from freezing the build.
RUN apt-get ... apache2 mariadb-server installs Apache (the web server) and MariaDB (the DB server).
EXPOSE 80 3306 lists the web port and the database port.
CMD starts MariaDB, then runs Apache, whenever the container starts.
4.3 Show the file.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ cat > Dockerfile <<'EOF'
FROM ubuntu:22.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get install -y apache2 mariadb-server && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80 3306

CMD service mariadb start && apache2ctl -D FOREGROUND
EOF
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ cat Dockerfile
FROM ubuntu:22.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get install -y apache2 mariadb-server && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80 3306

CMD service mariadb start && apache2ctl -D FOREGROUND
```

Step 5: Install and build the Dockerfile using Ansible

The playbook runs on localhost. Ansible copies the Dockerfile to a build folder and builds the image on this same workstation.

5.1 Install Ansible and the Python Docker library. The library lets Ansible control Docker.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ sudo apt install -y ansible python3-docker
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ansible is already the newest version (9.2.0+dfsg-0ubuntu5).
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  python3-docker python3-websocket
0 upgraded, 2 newly installed, 0 to remove and 142 not upgraded.
Need to get 127 kB of archives.
After this operation, 781 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-websocket all 1.7.0-1 [38.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-docker all 5.0.3-1ubuntu1.1 [89.1 kB]
Fetched 127 kB in 2s (55.4 kB/s)
Selecting previously unselected package python3-websocket.
(Reading database… 224625 files and directories currently installed.)
Preparing to unpack …/python3-websocket_1.7.0-1_all.deb…
Unpacking python3-websocket (1.7.0-1)…
Selecting previously unselected package python3-docker.
Preparing to unpack …/python3-docker_5.0.3-1ubuntu1.1_all.deb…
Unpacking python3-docker (5.0.3-1ubuntu1.1)…
Setting up python3-websocket (1.7.0-1)…
Setting up python3-docker (5.0.3-1ubuntu1.1)…
```

5.2 Install the Ansible Docker collection. It has the module that builds images.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ ansible-galaxy collection install community.docker
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to reinstall them, consider using `--force`.
```

5.3 Create the inventory. It only lists the workstation itself.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ cat > inventory.ini <<'EOF'
[local]
localhost ansible_connection=local
EOF
```

ansible_connection=local tells Ansible to run on this machine without SSH.

5.4 Create the playbook.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ cat > build_docker.yml <<'EOF'
- name: Build the web and database Docker image locally
  hosts: localhost
  tasks:
    - name: Create the Docker build directory
      file:
        path: "{{ ansible_env.HOME }}/web-db-build"
        state: directory

    - name: Copy the Dockerfile
      copy:
        src: Dockerfile
        dest: "{{ ansible_env.HOME }}/web-db-build/Dockerfile"

    - name: Build the web and database image
      community.docker.docker_image:
        name: web-db-server
        tag: latest
        source: build
        build:
          path: "{{ ansible_env.HOME }}/web-db-build"

    - name: List Docker images
      command: docker images
      register: docker_images
      changed_when: false

    - name: Display Docker images
      debug:
        var: docker_images.stdout_lines
EOF
```

What each task does:
Create the Docker build directory: makes web-db-build in your home folder.
Copy the Dockerfile: puts your Dockerfile in that folder.
Build the web and database image: builds an image named web-db-server from it.
List Docker images: runs docker images.
Display Docker images: prints the list so it shows in the output.

5.5 Run the playbook.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ ansible-playbook -i inventory.ini build_docker.yml



PLAY [Build the web and database Docker image locally] ************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [localhost]

TASK [Create the Docker build directory] **************************************************************************
changed: [localhost]

TASK [Copy the Dockerfile] ****************************************************************************************
changed: [localhost]

TASK [Build the web and database image] ***************************************************************************
changed: [localhost]

TASK [List Docker images] *****************************************************************************************
ok: [localhost]

TASK [Display Docker images] **************************************************************************************
ok: [localhost] => {
    "docker_images.stdout_lines": [
        "IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA",
        "hello-world:latest    5e2309035332       25.9kB         9.49kB   U    ",
        "ubuntu:22.04          5ec03bb3441e        119MB          31.7MB        ",
        "web-db-server:latest  b655fb10c94e        618MB           117MB        "
    ]
}

PLAY RECAP ********************************************************************************************************
localhost                  : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

5.6 Optional check: run the image and open the web server.

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ docker run -d --name webdb-test -p 8080:80 web-db-server:
latest
82df05f1b00026eb006ebae1345818c50cd0e42238ae57a98ffda758c9ad5641
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ curl -I http://localhost:8080
HTTP/1.1 200 OK
Date: Mon, 12 Oct 2026 12:37:52 GMT
Server: Apache/2.4.52 (Ubuntu)
Last-Modified: Mon, 12 Oct 2026 12:32:10 GMT
ETag: "29af-65da3e303e680"
Accept-Ranges: bytes
Content-Length: 10671
Vary: Accept-Encoding
Content-Type: text/html

wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ docker rm -f webdb-test
webdb-test
```

Step 6: Add, commit and push it to your repository

```
wdjavillonar@workstation:~/CPE212_Javillonar/HOA11.1_Docker_Web_DB$ cd ~/CPE212_Javillonar
wdjavillonar@workstation:~/CPE212_Javillonar$ git add HOA11.1_Docker_Web_DB
wdjavillonar@workstation:~/CPE212_Javillonar$ git commit -m "Add HOA11.1 containerization with Docker and Ansible"
[main 1d8cbaa] Add HOA11.1 containerization with Docker and Ansible
 3 files changed, 42 insertions(+)
 create mode 100644 HOA11.1_Docker_Web_DB/Dockerfile
 create mode 100644 HOA11.1_Docker_Web_DB/build_docker.yml
 create mode 100644 HOA11.1_Docker_Web_DB/inventory.ini
wdjavillonar@workstation:~/CPE212_Javillonar$ git push origin main
Username for 'https://github.com': MadhelKathlynAnnJavillonar
Password for 'https://MadhelKathlynAnnJavillonar@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 997 bytes | 997.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/MadhelKathlynAnnJavillonar/CPE212_Javillonar.git
   f9b7863..1d8cbaa  main -> main
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:
1. What are the benefits of implementing containerizations?

Containerization improves resource utilization, application consistency, and deployment efficiency. It enables applications to run consistently across different computing environments while using fewer system resources than traditional virtual machines. It also simplifies application management by making deployment and maintenance more efficient.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Helping with the commands, organizing step by step and used for grammar correction.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I confirm that all of the ideas are mine and did not replace my own original thinking.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.



I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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